

ALKANES

The simplest organic compound which are composed of carbon and hydrogen only are known as hydrocarbon. Hydrocarbons of two types:-

(1) saturated hydrocarbons (Alkane)

(2) Unsaturated hydrocarbons (Alkene and Alkyne)

* What are Alkane :-

The open chain hydrocarbon in which carbon atoms are linked with one another as well as hydrogen atom. By single or sigma mode. are known as alkanes.

Alkanes are known as paraffins which means with no affinity because these hydrocarbons are not to react with any common reagent like sulphuric acid (H_2SO_4), HNO_3 , Bromine water, KMnO_4 , Kolt etc.

Alkanes show generally, Substitution Reaction.

The general Formula of Alkane is $\text{C}_n\text{H}_{2n+2}$.

where, $n=1$, CH_4 methane

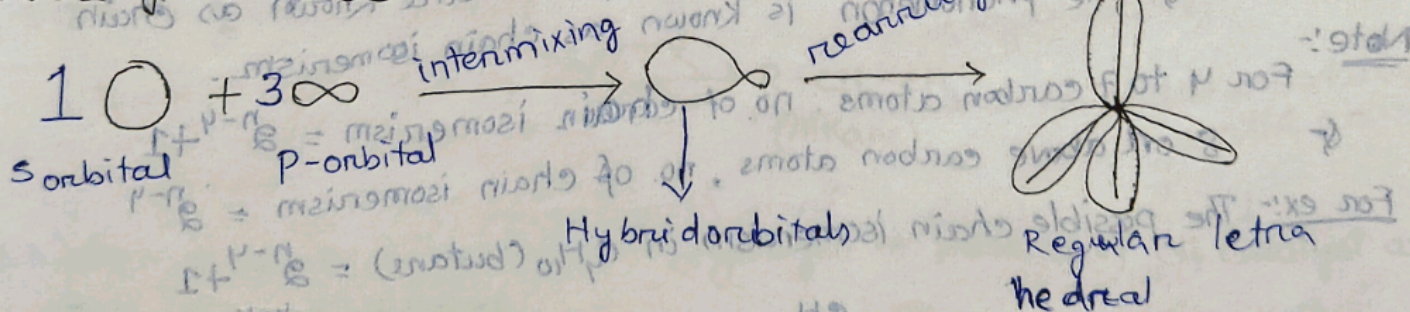
$n=2$, C_2H_6 / $\text{H}_3\text{C}-\text{CH}_3$ Ethane

$n=3$, C_3H_8 / $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_3$ Propane

$n=4$, C_4H_{10} / $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$ Butane.

All alkanes are sp^3 hybridisation, Bond angle is 109.5° and shape is Regular tetrahedron.

* sp^3 hybridisation :-

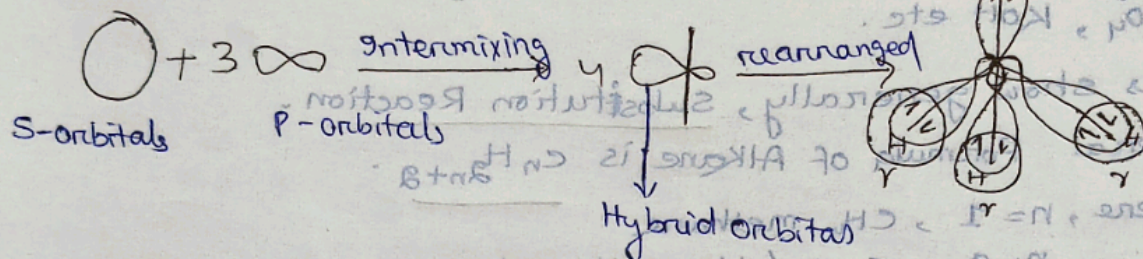
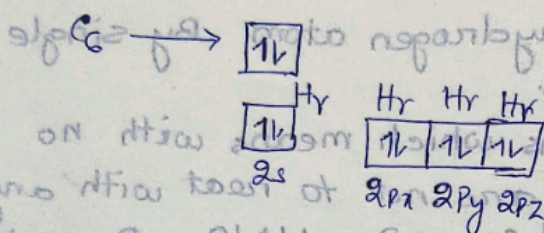


When one S-orbital and three P-orbitals intermixing with each other to give 4sp^3 hybrid orbitals. This phenomenon is known as sp^3 -hybridisation.

* Shape of Methane molecule:-

In CH_4 the central atom is carbon having atomic number 6 and has the electronic configuration in the excited state is $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$

Here, one $2s$ and three ($2p_x, 2p_y, 2p_z$) orbitals intermixing with each other to give $4sp^3$ hybrid orbitals. Each sp^3 hybrid orbital has 1 electron and has intermix with $1s$ atomic orbital of 4 H-atoms to form 4 (C-H), σ -Bonds. Due to sp^3 hybridisation the shape of the molecule is regular tetrahedron and Bond angle is $109^\circ 28' / 109.5^\circ$.



* Chain isomerism:-

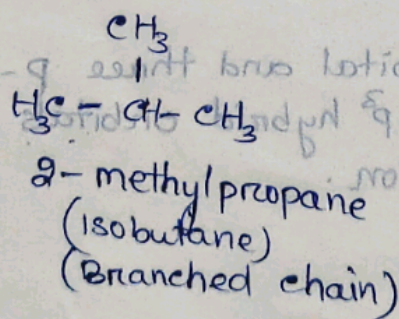
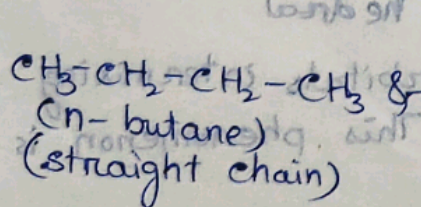
When the same molecular formula represents two or more compounds which differ in the nature of carbon chain i.e. $\rightarrow$ straight or branched, then such compounds are known as chain isomers & the phenomenon is known as chain isomerism.

Note:-

For 4 to 7 carbon atoms, no. of chain isomerism = $2^{n-4} + 1$

& 8 or above carbon atoms, no. of chain isomerism = 2^{n-4}

For ex:- The possible chain isomerism in C_4H_{10} (butane) = $2^{4-4} + 1$



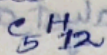
$$= 2^{4-4} + 1$$

$$= 2^0 + 1$$

$$= 1 + 1$$

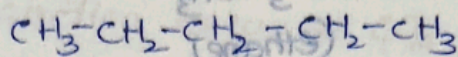
$$= 2$$

Ex:-2

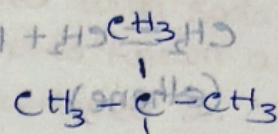


C_5H_{12} represent 3 chain Isomers.

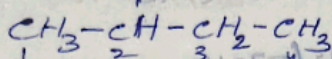
$$\begin{aligned} & 2^{n-4} + 1 \\ & = 2^{5-4} + 1 \\ & = 2^1 + 1 \\ & = 2 + 1 \\ & = 3 \end{aligned}$$



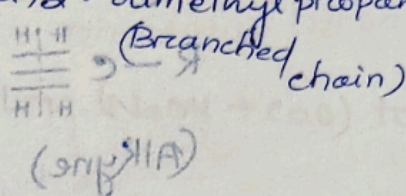
(Pentane)
(straight line)



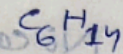
(Neopentane)
(2,2-dimethylpropane)



2-methylbutane
(Branched chain)

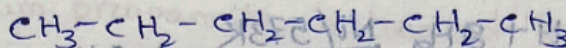


Ex:-3

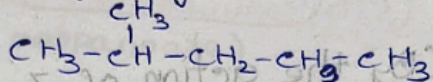


C_6H_{14} represent 5 chain Isomers.

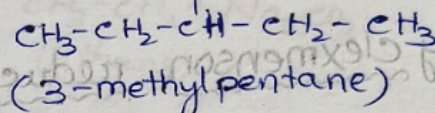
$$\begin{aligned} & = 2^{6-4} + 1 \\ & = 2^2 + 1 \\ & = 4 + 1 \\ & = 5 \end{aligned}$$



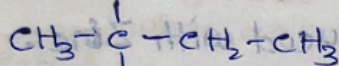
(n-Hexane)
(straight chain)



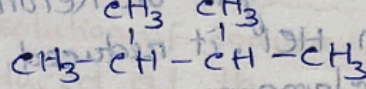
(2-methylpentane)



(3-methylpentane)



(2,2-Dimethylbutane)

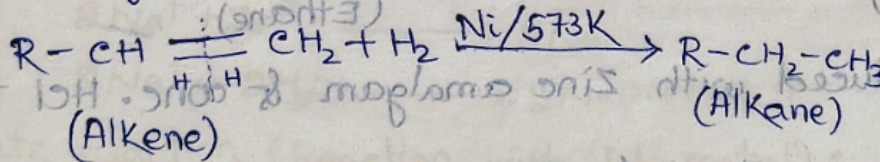


(2,3-Dimethylbutane)

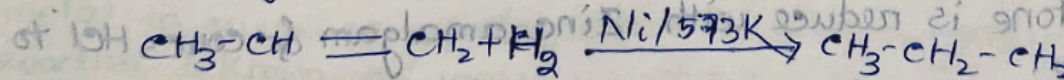
* Method of preparation :-

1. By reduction of Alkene & Alkyne (Sabatier - Senderens reaction) :-

(a) When an alkene is reduced with H_2 in presence of Ni catalyst at 573K to give an alkane.



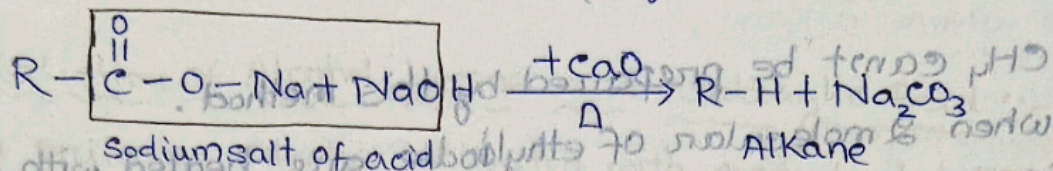
Ex:- when propene is added with hydrogen in presence of Ni catalyst at 573K to give propane.



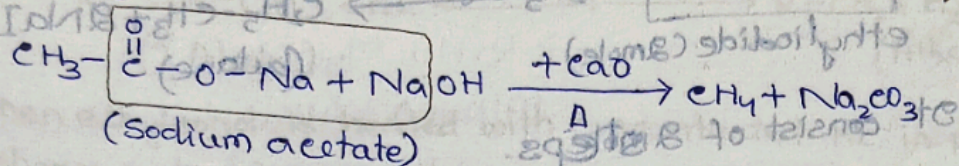
(propene)

(propane)

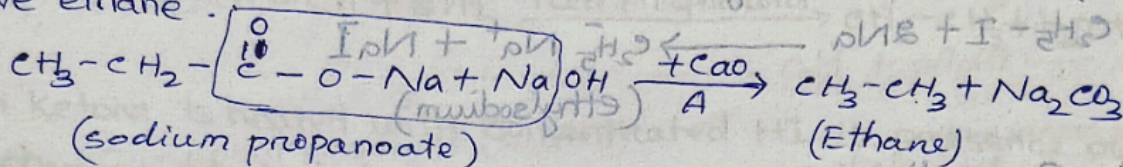
③ By Decarboxylation of sodium salt of carboxylic acid (Duma's method) :-
 when a sodium salt of carboxylic acid is heated with sodalime ($\text{NaOH} + \text{CaO}$) to give an Alkane. This reaction is known as Decarboxylation reaction because CO_2 gas is removed as Na_2CO_3 .



Ex:- when sodium acetate is heated with sodalime ($\text{NaOH} + \text{CaO}$) to give methane.

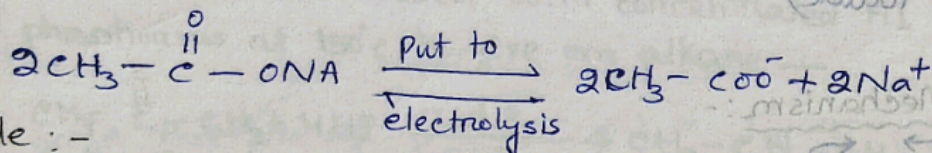


Ex:- when sodium propanoate is heated with sodalime ($\text{NaOH} + \text{CaO}$) to give ethane.

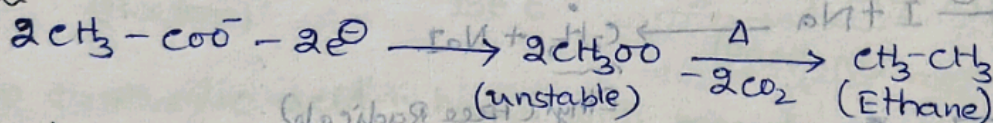


4. Kolbe's Synthesis (Kolbe's electrolytic method) :-

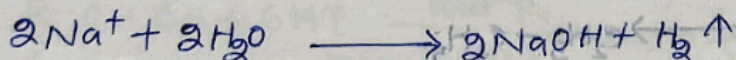
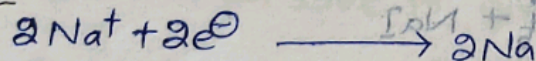
when an aqueous solution of sodium acetate is put to electrolysis, ethane is produced at the anode & H_2 gas is liberated at the cathode.



At anode :-

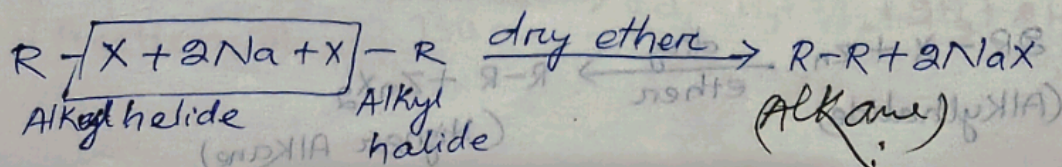


At cathode :-



5. Wurtz reaction (Reaction with Na metal) :-

This method is used to produce symmetrical alkane. when two molecules of Alkyl halide are heated with sodium metal in presence of dry ether to give symmetrical alkane. This reaction is known as Wurtz reaction.



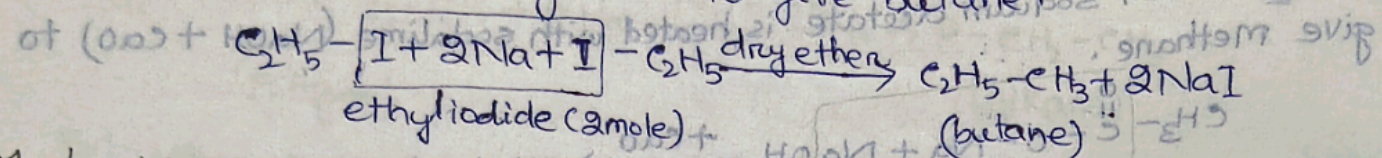
when 2 molecules of methyl iodide are heated with sodium metal in presence of dry ether to give ethane.

$$CH_3-I + 2Na + I-CH_3 \xrightarrow{\text{dry ether}} CH_3-CH_3 + 2NaI$$

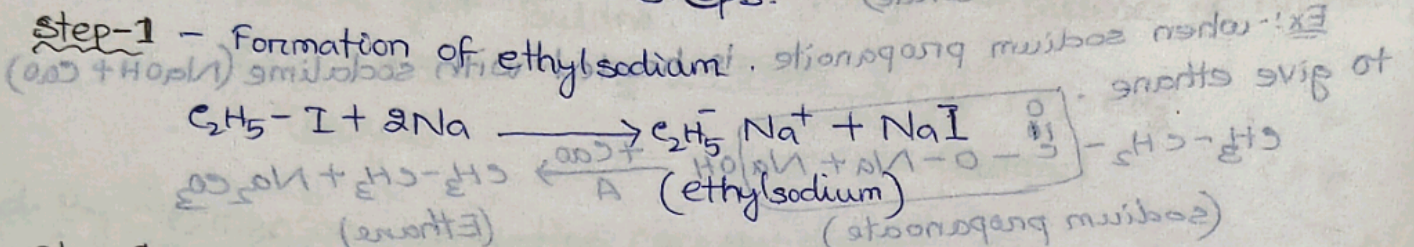
methyl iodide (2mole)

* Note :- CH_4 can't be prepared by this method.

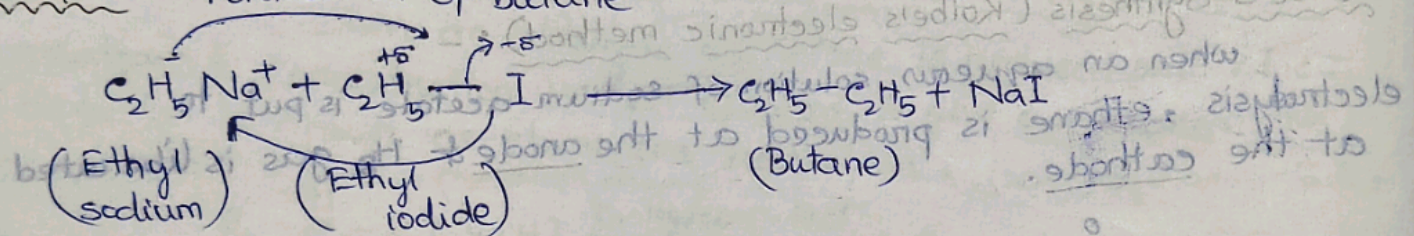
when 2 molecular of ethyl iodide are heated with sodium metal in presence of dry ether to give butane.



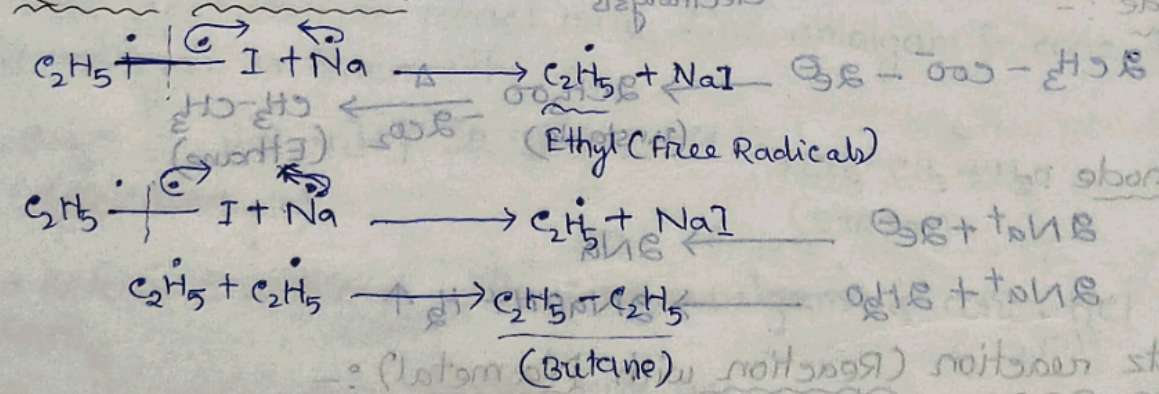
Mechanism :- It consist of 2 steps.



Step-2 :- Formation of butane

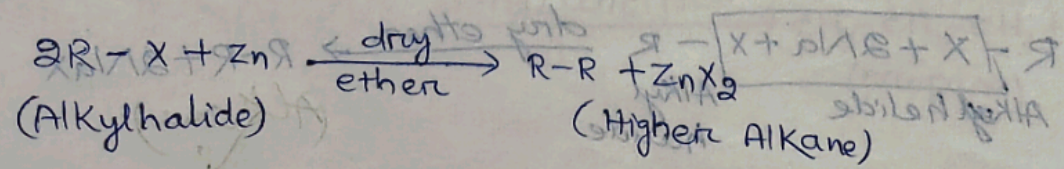


② Free Radicals Mechanism :-



③ Reaction with Zn metal (Frankland's Reaction) :-

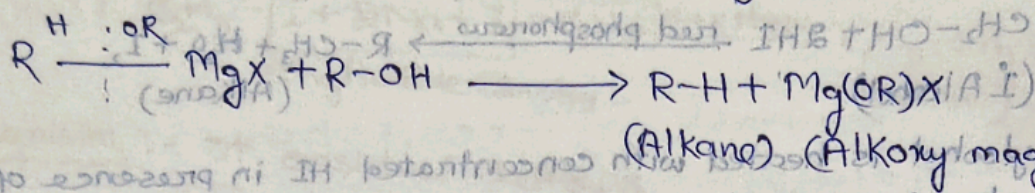
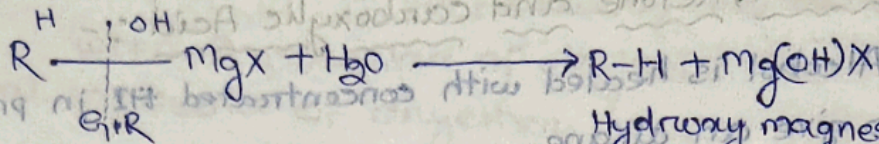
when An Alkyl halide is heated with Zn metal in presence of dry ether to give higher Alkane. This reaction is known as Frankland's Reaction.



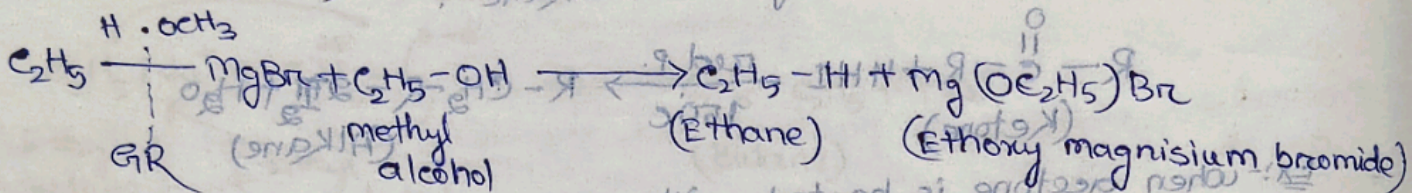
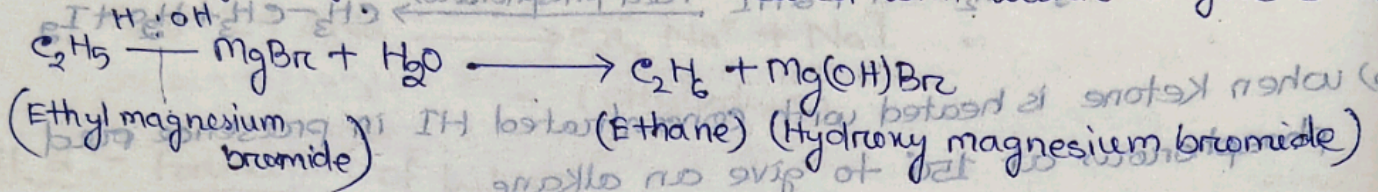
8. By hydrolysis of Grignard reagent ($R-MgX$):

Alkyl magnesium halide is known as Grignard reagent ($R-MgX$)

when Alkyl magnesium halide is prepared with water or alcohol to give alkane.



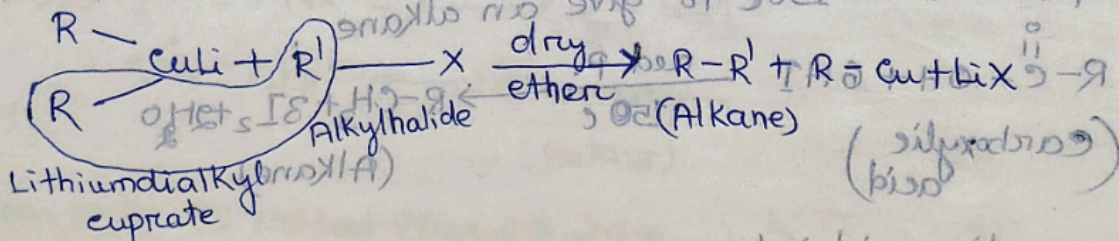
Ex:- when ethyl magnesium bromide is heated with water to give ethane



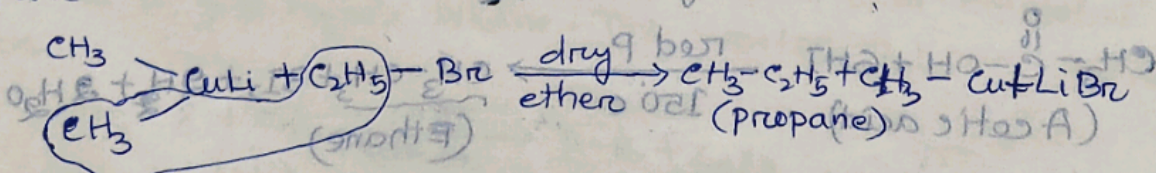
9. By Corey-House synthesis:

This synthesis is used for preparing both symmetrical as well as unsymmetrical alkane.

when alkyl halide is react with lithium dialkyl cuprate (R_2CuLi) in presence of dry ether to give Alkanes

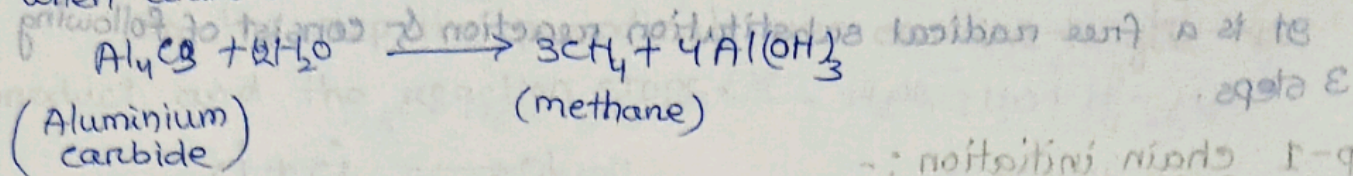


Ex:- when Lithium dimethyl cuprate react with ethyl bromide to give propane.



Note:-

when aluminium carbide is treated with water to give methane



* Physical properties :-

- ① The first four members of alkanes C_1 to C_4 are colourless gases with petrol like smell, The next 13 (C_5 to C_{17}) are colourless liquid and higher alkanes are colourless solid.
- ② Alkanes are non-polar having dipolar moment $\mu = 0$ Debye

$$= 0 \text{ D}$$

$1 \text{ Debye} = 10^{-18} \text{ e.s.u} \times \text{cm}$

 and hence they are insoluble in water but soluble in non-polar solvent like Benzene, ether, CCl_4 etc.
- ③ The Boiling point of alkanes increases with increase in their molecular masses but the point of branched chain isomer lower than straight chain isomer.
- ④ Lower alkanes have low density due to low molecular mass but higher alkanes have greater density mass per volume as $\rho = \frac{m}{V}$.

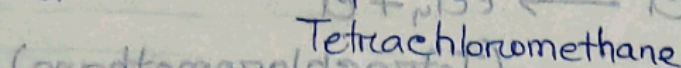
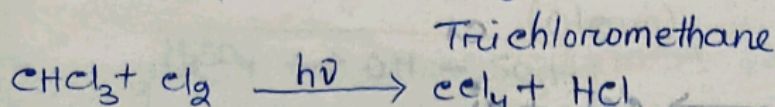
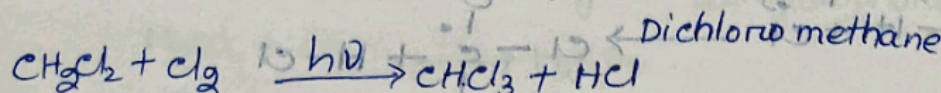
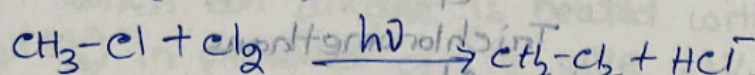
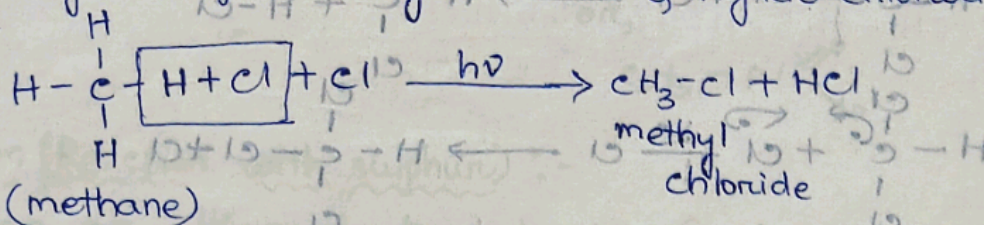
* Chemical properties :-

Alkanes shows free radical substitution reaction :-

(a) Chlorination (Halogenation) :-

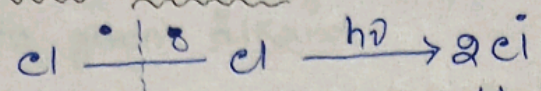
when methane react with chlorine in presence of U.V light or diffuse sunlight to give methyl chloride & Higher chloroderivation.

Mechanism :-



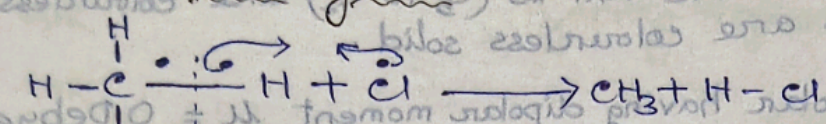
* Mechanism - It is a free radical substitution reaction & consist of following 3 steps.

Step-1 chain initiation :-



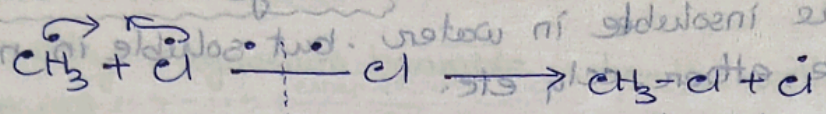
chlorine free radicals.

Step-2 chain propagation



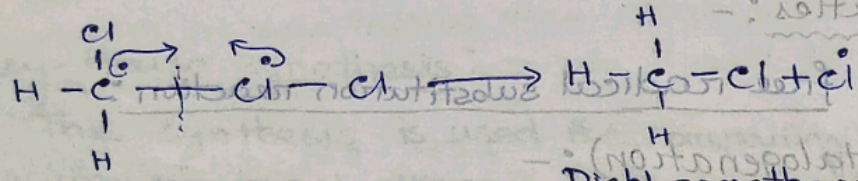
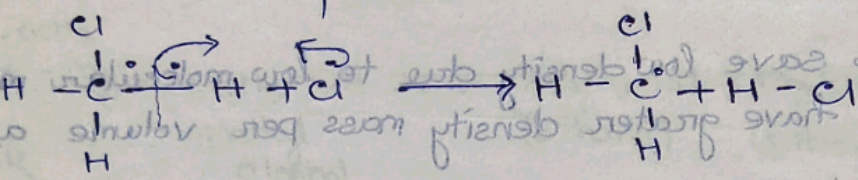
(Methane)

methyl free radical

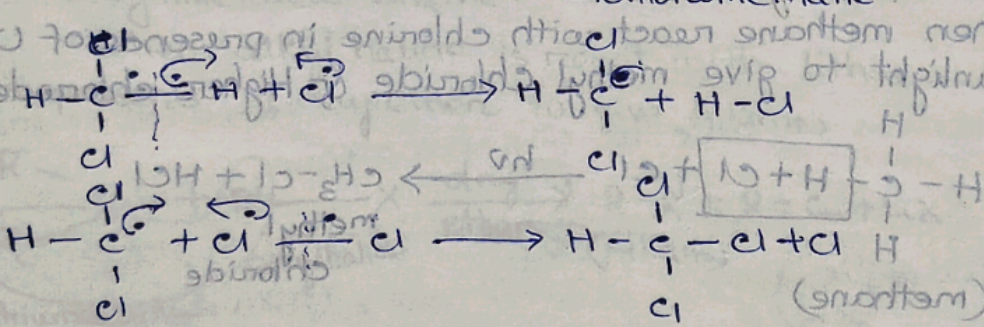


methyl chloride

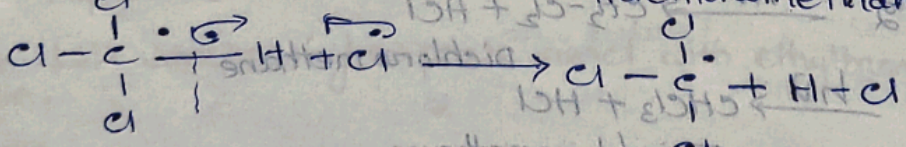
The above two steps repeated again and again to give a final product CCl_4



Dichloromethane



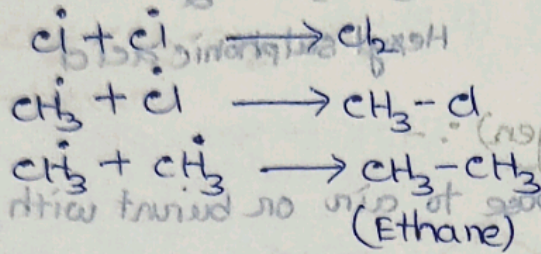
Trichloromethane



(Tetrachloromethane)

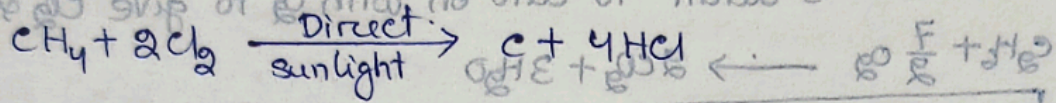
Step 3 of chain termination

On the step the periodicals are combine to give new product and the reaction stops



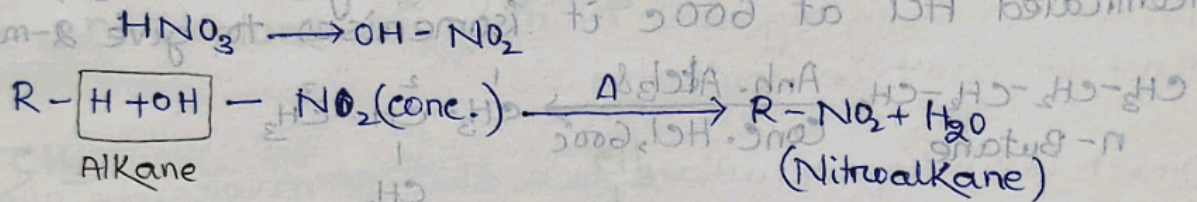
* Note :-

① when methane react with chlorine in presence of direct sunlight to give carbon and HCl.

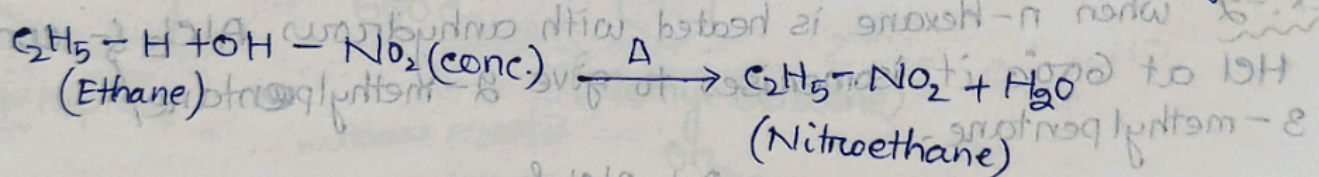


② Nitration (Reaction with Nitric acid (HNO₃)) :-

when alkane is heated with concentrated nitric acid to give nitroalkane & water.

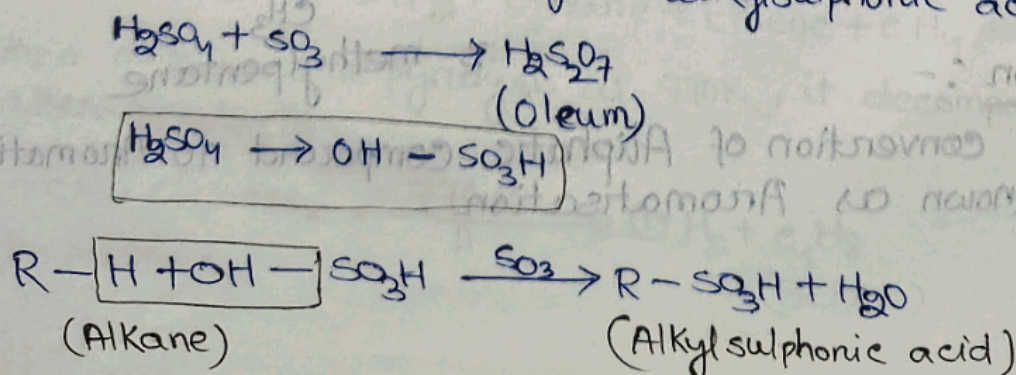


Ex:- when ethane is heated with concentrated nitric acid to give nitroethane.

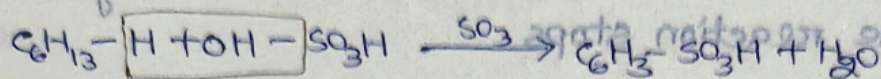


3. Sulphonation (Reaction with sulphur) :-

when an alkane is heated with fuming sulphuric acid (H₂SO₄ + SO₃) in presence of to give alkylsulphonic acid.



Ex:- when n-hexane is heated with fuming sulphuric acid to give hexylsulphonic acid

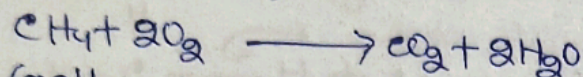


n-Hexane

Hexyl sulphonic acid

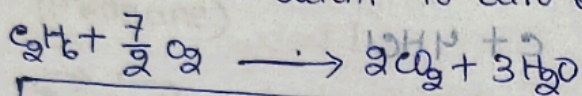
4. Combustion (Reaction with oxygen):

when methane is exposed to air or burnt with O_2 to give CO_2 and water (H_2O).

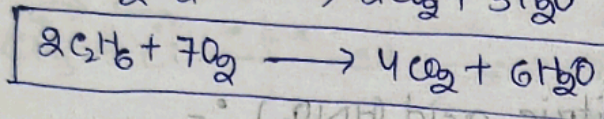


(methane)

Ex:- when ethane is burnt to air on with O_2 to give CO_2 & H_2O .

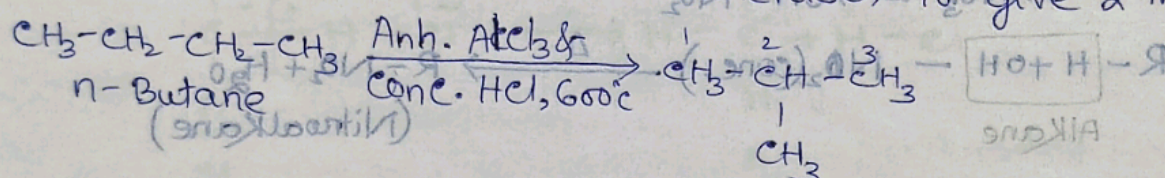


or,



5. Isomerisation:

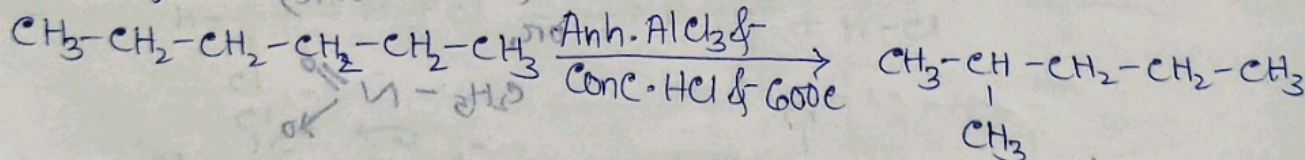
when n-Butane is heated with anhydrous AlCl_3 & concentrated HCl at 600°C it isomerises to give 2-methylpropane



n-Butane

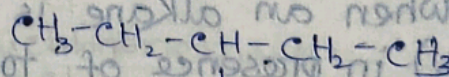
2-methylpropane

Ex:-2 when n-Hexane is heated with anhydrous AlCl_3 & conc. HCl at 600°C it isomerises to give 2-methylpentane / 3-methylpentane



CH₃

2-methylpentane

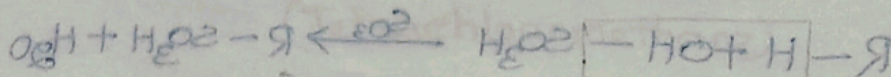


CH₃

3-methylpentane

6. Aromatisation:

The conversion of Aliphatic compound to Aromatic compound is known as Aromatisation.



(Alkyl sulphonic acid)

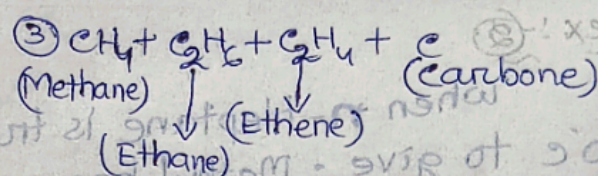
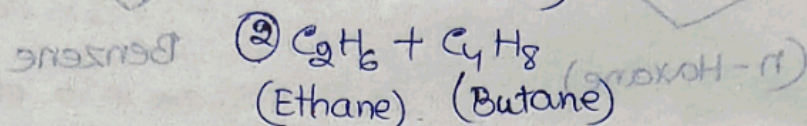
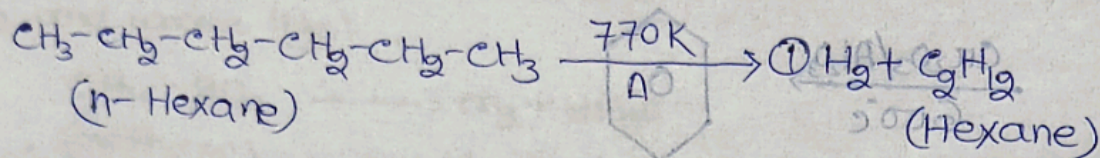
(Alkane)

For ex:- ① when n-Hexane is pyrolysis at 770K it decomposed into following smaller products.

② $C_2H_6 + C_2H_4$ (Ethane) (Ethene)

③ $CH_4 + C_2H_6$ (Methane) (Propane)

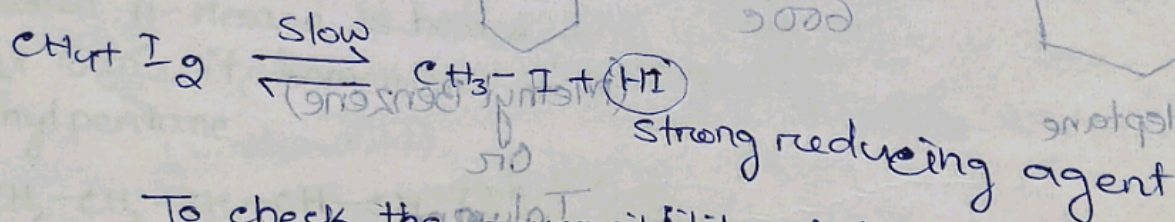
Ex:-2 when n-Hexane is pyrolysis at 770K it decomposed into following smaller product.



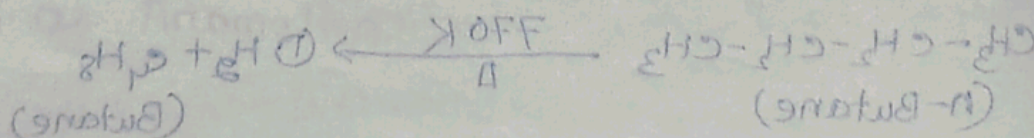
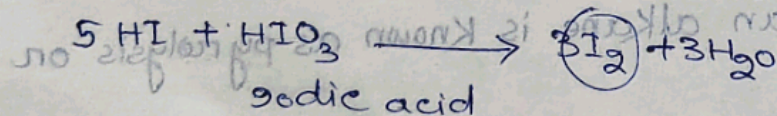
* Note :-

① Direct fluorination of alkane can't takes place because F_2 is an important oxidising agent & it makes the reaction violent & Hence it can't be controlled.

② Direct iodination of alkane is a slow & reversible process and the formation of HI is a strong reducing agent & it reduces to give methane.



To check the reversibility of the reaction, some oxidising agent like Nitric acid, Iodic acid etc added to the reaction mixture & to liberate free iodine



* Uses :-

- ① Naturally occurring alkenes in the form of natural gas and Petroleum product are commonly use as fuel.
- ② LPG is a mixture of butane & isobutane & commonly used as fuels.
- ③ Paraphine - wax is use as an electric insulator and for making both polishes.
- ④ These are use as solvent for organic for compounds.

* Structure of ethene molecule

Structure of ethene molecule

